```
let applications = [];
let idCounter = 1000;

function generateAppID() {
  return "APP" + idCounter++;
}

function submitApplication(name, email) {
  try {
    if (!/^[a-zA-Z\s]+$/.test(name)) {
      throw new Error("Invalid name: only letters and spaces allowed.");
    }

    if (!/^[\w.-]+@[a-zA-Z_]+?\.[a-zA-Z]{2,3}$/.test(email)) {
      throw new Error("Invalid email format.");
    }

    const application = {
      id: generateAppID(),
      name,
      email,
      status: "Pending",
    };

    applications.push(application);
    console.log(`Application submitted for ${name}. ID:
${application.id}`);
  } catch (err) {
    console.log("Error:", err.message);
  }
}

function checkStatus(application) {
  return new Promise((resolve) => {
    setTimeout(() => {
      const statuses = ["Reviewed", "Selected", "Rejected"];
      let result = statuses[Math.floor(Math.random() * statuses.length)];
      application.status = result;
      resolve(application);
    }, 1000);
  });
}

async function showAllStatuses() {
  console.log("\nChecking all applications...");
  for await (const [index, app] of applications.entries()) {
    await checkStatus(app);
    console.log(`${index + 1}. ${app.name} (${app.id}) - ${app.status}`);
  }
}

const nameIterator = {
  [Symbol.iterator]: function () {
    let index = 0;
    return {
      next: () => {
        if (index < applications.length) {
          return { value: applications[index++].name, done: false };
```

```
        } else {
          return { done: true };
        }
      },
    };
  },
};

submitApplication("Alice Johnson", "alice@mail.com");
submitApplication("Bob123", "bob@mail.com");
submitApplication("Charlie", "charlie.gmail.com");
submitApplication("David Lee", "david@company.com");

setTimeout(async () => {
  await showAllStatuses();

  console.log("\n👥 Applicants:");
  for (let name of nameIterator) {
    console.log("- " + name);
  }
}, 2000);
```